33) $5h^3+31h^2-28h$

$h(5h^2+31h-28)$

$5(-28)=-140$

$h(5h^2+35h-4h-28)$

$h\,5h(h+7)-4h+7)$

$h(5h-4)(h+7)$

35) $28k^2-260k+288$

$4(7k^2-65k+72$

$7(72)=504$

$4(7k^2-56k-9k+72)$

$4[7k(k-8)-9(k-8)]$

$4(7k-9)(k-8)$

37) $7p^2-31p+30$

$7(30)=210$

$7p^2-21p-10p+30$

$7p(p-3)-10(p-3)$

$(7p-10)(p-3)$

40) $7h^2+67h-30$

$7(-30)=-210$

$7h^2+70h-3h-30$

$7h(h+10)-3(h+10)$

$(7h-3)(h+10)$

34) $42h^2+180h+162$

$6(7h^2+30h+27)$

$7(27)=189$

$6(7h^2+21h+9h+27)$

$6(7h(h+3)+9(h+3))$

$6(7h+9)(h+3)$

36) $5h^2+16h+12$

$5h^2+16h+12$

$5(12)=60$

$5h^2+10h+6h+12$

$5h(h+2)+6(h+2)$

$(5h+6)(h+2)$

38) $7x^2+5x$

$x(7x+5)$

39) $42x^2+90x-108$

$6(7x^2+15x-18$

$7(-18)=-126$

$6(7x^2+21x-6x-18)$

$6[7x(x+3)-6(x+3)]$

$6(7x-6)(x+3)$

41) $24h^2-24h+6$

$6(4h^2-4h+1)$

$4h^2=(2h)^2,\ 1=1^2$

$-2(2h)(1)=-4h$

$6(2h-1)^2$

$6(2h-1)^2$

42) $24x^2+216x+486$

$6(4x^2+36x+81)$

$4x^2=(2x)^2,\ 81=9^2$

$2(2x)(9)=36x$

$\underline{6(2x+9)^2}$

43) $64m^2-112m+49$

$64m^2=(8m)^2,\ 49=7^2$

$-2(8m)(7)=-112m$

$\underline{(8m-7)^2}$

44) $100h^2+20h+1$

$100h^2=(10h)^2,\ 1=1^2$

$2(10)(1)=20h$

$\underline{(10h+1)^2}$

45) $81x^6-90x^3+25$

$81x^6=(9x^3)^2,\ 25=5^2$

$-2(9x^3)(5)=-90x^3$

$\underline{(9x^3-5)^2}$

46) $147k^6-336k^3+192$

$3(49k^6-112k^3+64)$

$49k^6=(7k^3)^2,\ 64=8^2$

$-2(7k^3)(8)=-112k^3$

$\underline{3(7k^3-8)^2}$

47) $36q^8-192q^5+256q^2$

$4q^2(9q^6-48q^3+64)$

$9q^6=(3q^3)^2,\ 64=8^2$

$-2(3q^3)(8)=-48q^3$

$\underline{4q^2(3q^3-8)^2}$

48) $49+28p^3+4p^6$

$4p^6+28p^3+49$

$4p^6=(2p^3)^2,\ 49=7^2$

$2(2p^3)(7)=28p^3$

$\underline{(2p^3+7)^2}$

49) $9x^2-49$

$9x^2=(3x)^2,\ 49=7^2$

$a^2-b^2=(a-b)(a+b)$

$\underline{(3x-7)(3x+7)}$

50) $4q^2-81$

$4q^2=(2q)^2,\ 81=9^2$

$\underline{(2q-9)(2q+9)}$

51) $27-147p^2$

$3(9-49p^2)$

$9=3^2,\ 49p^2=(7p)^2$

$\underline{3(3-7p)(3+7p)}$

52) $250h^2 - 490$

$10(25h^2 - 49)$

$25h^2 = (5h)^2, \quad 49 = 7^2$

$10(5h-7)(5h+7)$

53) $81k^6 - 16$

$81k^6 = (9k^3)^2, \quad 16 = 4^2$

$(9k^3-4)(9k^3+4)$

54) $x^6 - 9$

$x^6 = (x^3)^2, \quad 9 = 3^2$

$(x^3-3)(x^3+3)$

55) $25h^6 - 49$

$25h^6 = (5h^3)^2, \quad 49 = 7^2$

$(5h^3-7)(5h^3+7)$

56) $600q^7 - 6q$

$6q(100q^6 - 1)$

$100q^6 = (10q^3)^2, \quad 1 = 1^2$

$6q(10q^3-1)(10q^3+1)$